

REASONING BASED DI

Directions for questions 1 to 5: Answer these questions on the basis of the information given below.

The table gives the number of people getting in and getting down of a local train which starts at station A and goes to station F, stopping at stations B, C, D and E in between. The first set of passengers got in at station A and all the passengers got down at F, the last station.

Station	Getting in	Getting down
A	28	–
B	21	11
C	29	23
D	26	18
E	17	31
F	–	

At each station, the first person to get in, gets in after the last person to get down at that station, gets out.

What is the number of passengers who got down at station F?

☐ a) 41

☒ b) 38 (Your answer - ✓)

☐ c) 32

☐ d) 27

Solution:

As all passengers get down at the last station, the number of passengers getting down would be $(28 + 21 + 29 + 26 + 17) - (11 + 23 + 18 + 31)$
 $= 121 - 83$
 $= 38.$

Choice (B)

Directions for questions 2 and 3: Type in your answer in the input box provided below the question.

At least how many people got down at the next station after the station they got in?

Your Answer: 42 ✓

Solution:

All the eleven people who got down at station B had got in at the previous station. Similarly, at least $6(23 + 11 - 28)$ people who got down at station C got in at station B. In the same way, three people who got in at C got down at station D, at least five who got in at station D got down at E and all 17 who got in at E got down at F.
 $\therefore$ at least $11 + 6 + 3 + 5 + 17 = 42$ people got down at the next station after the station they got in.
Ans: (42)

Directions for questions 2 and 3: Type in your answer in the input box provided below the question.

At least how many people got down two stations after the station they got in?

Your Answer: 2 ✓

Solution:

At least two people who got in at A got down at C. The 21 people who got in at B could have got down at C. Among the 18 people who got down at D, 10 could have got in at station A and eight could have got in at station C. Out of the 31 people who got down at station E, 26 could have got in at D and the remaining five at station A. The 38 who got down at station F would be from E and C (21). So at least two people got down two stations after the station they got in.
Ans: (2)

Directions for questions 4 and 5: Select the correct alternative from the given choices.

At most how many people travelled from the first station to the last station?

☐ a) 17

☐ b) 13

☒ c) 11 (Your answer - ✕)

☐ d) 15

Solution:

Among the people who got in at station A, eleven got down at station B and at least two got down at station C. The remaining 15 people could have travelled to the last station.
Choice (D)

Directions for questions 4 and 5: Select the correct alternative from the given choices.

At most how many people could have got in at station B and got down at station E?

☐ a) 15 (Correct Answer - ✓)

☐ b) 17

☐ c) 19

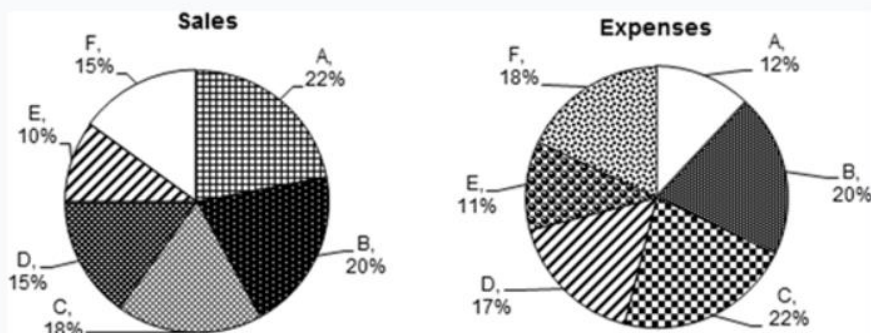
☐ d) 21

Solution:

At least six people who got in at B have to get down at C. The remaining 15 people could have got down at E.
Choice (A)

Directions for questions 6 to 10: Answer these questions on the basis of the information given below.

The following pie charts give the breakup of sales and expenses of six products A – through F – of a company PQR.



Profit = Sales – Expenses

Profit (%) =

$$\frac{\text{Profit}}{\text{Expenses}}$$

× 100

If it is known that the overall profit on the six products is 10%, on how many products did the company have more than 10% profit?

☐ a) 1 (Correct Answer - ✓)

☐ b) 2

☐ c) 3

☐ d) Cannot be determined

Solution:

Any product for which the percentage contribution in sales is more than its percentage contribution in expenses will have more than 10% profit, i.e., product A. Choice (A)

If the profit on product B was Rs.10cr, at least how many of the six products made a profit, if it is known that the total expenses were not more than 90% of the sales?

☐ a) 2

☐ b) 3 (Correct Answer - ✓)

☐ c) 1

☐ d) 4

Solution:

Let the total sales be x and the total expenses be y .

$$0.2x - 0.2y = 10 \text{ cr}$$

$$x - y = 50 \text{ cr}$$

As the expenses were not more than 90% of the sales, the maximum value of sales is 500 cr and that of expenses is 450 cr.

$\therefore$ the company made a profit on at least three products – A, B and E. Choice (B)

If the overall profit on the six products is Rs.120cr, how many products had a profit more than Rs.20cr?

- ☐ a) 1
- ☐ b) 2
- ☐ c) 3
- ☐ d) Cannot be determined

Solution:

Product A will definitely have a profit more than ₹20 cr. For product C, depending on the value of sales and expenses, the profit can be more or less than ₹20 cr. For example, if the sales are ₹130 cr and the expenses are ₹10 cr, the profit on product C is 21.2 cr. If the sales are ₹180 cr and the expenses are ₹60 cr, the profit on product C is 19.2 cr.
Choice (D)

If the profit percentage on product E is 10, what is the profit percentage on product B?

- ☐ a) 8
- ☐ b) 15
- ☐ c) 21
- ☐ d) 25

Solution:

Assume total expenses = 100
Expenses on E = 11
As profit percentage of E is 10, the sales of E = 12.1.
∴ total sales = 121.
AS B has 20% share in both sales and expenses, its profit percentage would be 21.
Choice (C)

If it is known that all the products made a profit, what is the minimum overall profit percentage on the six products?

- ☐ a) 20
- ☐ b) 22.2
- ☐ c) 24.6
- ☐ d) 28.1

Solution:

C is the product with the least profitability. If X and Y are the total sales and total expenses respectively,

$$0.18X > 0.22Y$$

or

$$\frac{X}{Y} > \frac{22}{18}$$

$$\therefore \text{the overall profit percentage is } \frac{22 - 18}{18} \times 100 = 22.2\%.$$

Choice (B)

Directions for questions 11 to 15: Answer these questions on the basis of the information given below.

The table below gives the range of marks scored in an exam by students in five classes.

Class	Marks (x)				
	$x \geq 90\%$	$x \geq 75\%$	$x \geq 60\%$	$x \geq 40\%$	$x < 40\%$
6	6	23	35	46	4
7	11	26	31	38	9
8	17	29	40	48	5
9	8	23	35	49	3
10	14	19	32	38	7

Students who score less than 40 percentage marks fail the exam.

If all students in each class wrote the exam, which class had the highest number of students?

- ☐ a) 6
- ☐ b) 7
- ☐ c) 8
- ☐ d) 9

(Correct Answer - ✓)

Solution:

The number of students in the different classes are as follows:

$$\text{Class 6} - 46 + 4 = 50$$

$$\text{Class 7} - 38 + 9 = 47$$

$$\text{Class 8} - 48 + 5 = 53$$

$$\text{Class 9} - 49 + 3 = 52$$

$$\text{Class 10} - 38 + 7 = 45$$

Class 8 has the highest number of students.

Choice (C)

If students who scored at least 60 but less than 90 percentage marks are awarded a first class, in which class is the number of students who scored a first class the highest?

☒ a) 6 (Correct Answer - ✓)

☐ b) 8

☐ c) 9

☐ d) 10

Solution:

The number of students who scored a first class in the different classes are

$$\text{Class 6} - 35 - 6 = 29$$

$$\text{Class 7} - 31 - 11 = 20$$

$$\text{Class 8} - 40 - 17 = 23$$

$$\text{Class 9} - 35 - 8 = 27$$

$$\text{Class 10} - 32 - 14 = 18$$

The highest is in Class 6.

Choice (A)

Any student scoring at least 40 but less than 60 percentage marks in the exam is awarded a second class. What is the total number of students who were awarded second class in the exam in the five classes together?

☐ a) 31

☐ b) 38

☐ c) 42

☒ d) None of these (Correct Answer - ✓)

Solution:

The students who were awarded a second class in the different classes are as follows:

$$\text{Class 6} - 46 - 35 = 11$$

$$\text{Class 7} - 38 - 31 = 7$$

$$\text{Class 8} - 48 - 40 = 8$$

$$\text{Class 9} - 49 - 35 = 14$$

$$\text{Class 10} - 38 - 32 = 6$$

$$\text{The required number} = 11 + 7 + 8 + 14 + 6 = 46.$$

Choice (D)

What percentage of the total students in the five classes together, failed in the exam?

☐ a) 9.6

☐ b) 10.2

☒ c) 11.3 (Correct Answer - ✓)

☐ d) 12.7

Solution:

The percentage of students who failed

$$= \frac{28}{50 + 47 + 53 + 52 + 45} \times 100$$

$$= \frac{28}{247} \times 100$$

$$= 11.33\%.$$

Choice (C)

What can be the maximum difference in the number of students who scored at least 60 but less than 80 percentage marks in any two of the five classes?

- ☐ a) 24 (Correct Answer - ✓)
- ☐ b) 22
- ☐ c) 19
- ☐ d) None of these

Solution:

The minimum and maximum values of the number of students who scored at least 60 percentage but less than 80 percentage marks in the different classes are as follows:

Class 6 – 12, 29

Class 7 – 5, 20

Class 8 – 11, 23

Class 9 – 12, 27

Class 10 – 13, 18

The difference would be maximum when maximum number of students score in that range in class 6 and the minimum number of students score in that range in class 7.

The difference = $29 - 5 = 24$.

Choice (A)